LANDSLIDE GEOHAZARD ASSESSMENT REPORT

Integrated Geodetic InSAR Diagnostics & Large Language Model Interpretability

Site Name:	Site
Diagnostic ID:	TEST001
Analysis Date:	2026-07-08
Geographic coordinates:	0°N, 0°E
Computed Risk Rating:	**HIGH**

Prepared by the AI-Powered Geohazard Analytics System

Table of Contents

1. Executive Summary

An integrated geotechnical hazard assessment has been compiled for the surveyed site. Leveraging spaceborne multi-temporal interferometric synthetic aperture radar (InSAR) datasets and SegFormer-B2 computer vision models, our system has mapped active surface displacement boundaries. The computed core sliding zone is estimated at **0.0 m²**. Due to the steep geomorphic configuration and high precipitation triggers, the slope is classified at a **HIGH** geohazard status, requesting prompt stabilization interventions.

2. Technical Analysis & Geodetic Metrics

The geodetic displacement vectors reveal downward line-of-sight (LOS) motion creep:

Interferometric Parameter	Monitored Value	Scientific Interpretation
Mean LOS Velocity	1 mm/yr	Average downslope creep trend
Peak LOS Velocity	2 mm/yr	Critical maximum displacement velocity
Mean Acceleration	0.1 mm/yr²	Indicates accelerating/decelerating slope phase
Radar Coherence Index	0.8	Correlation index measuring signal quality
Slope Angle / Aspect	30° / N	Gradient trigger values of monitored slide zone

3. Spatial Maps & Visual Results

The spatial distribution of active landslide boundaries compared across amplitude bands, prediction masks, heatmaps, and overlays:

IMAGE MAP NOT INGESTED
(Visual Pending)

Fig 1: Input InSAR Amplitude Map

IMAGE MAP NOT INGESTED
(Visual Pending)

Fig 2: SegFormer Prediction Mask

IMAGE MAP NOT INGESTED
(Visual Pending)

Fig 3: InSAR Displacement Heatmap

IMAGE MAP NOT INGESTED
(Visual Pending)

Fig 4: Blended Spatial Overlay

4. Emergency Recommendations & Mitigation Actions

- **Immediate Actions (24-48 Hours):** Restrict access to slope toe sectors and implement community warning networks.
- **Short-Term (1-4 Weeks):** Install inclinometers and extensometer sensor grids to log displacement speeds.
- **Long-Term:** Construct retaining shear-key walls and lay horizontal drainage lines.

5. Plain-Language Summary

Satellite measurements show that the slope at the surveyed site is shifting downwards. It is slowly slipping at a rate of 1 mm per year. While this creep is slow, when combined with the steep slope angle, it creates a very high risk of a sudden landslide during heavy rainfall. Residents should look for ground cracks or leaning trees and stay alert for emergency notices.

6. Appendix & Geotechnical Metadata

This assessment is generated through the combination of interferometric satellite sensors and deep-learning report compilers.

Metadata Field	Value / Configuration	System Impact
SegFormer Backing	nvidia/segformer-b2-finetuned-ade	Active landslide body segmentation
Analysis Engine	Geodetic Extraction V1.4	Extracts area, coordinates, aspect, and slope stats
VLM Framework	Modular VLM Wrapper	Generates report text using image+text models
Failure Probability Score	0.4	Computed using gradient & displacement ratio
Signal Coherence	0.8	Measures phase quality of the radar backscatter

Prepared By:

Geohazard AI Engine

Approved By:

Lead Geotechnical Specialist